

ONLINE APPENDICES

Appendix A: Positioning Relative to the Three Most Closely Related Studies

Table A. Positioning Relative to the Three Most Closely Related Studies

Dimension	Xue et al. (2026)	Shan & Qiu (2026)	Su et al. (2025)	Our study
Primary outcome	Question (supply & quality)	Answer (supply, length & readability)	Answer (supply & length)	Answer (supply & quality)
Domain and basis of contribution value	Technical; correctness	Technical; correctness	General-interest; experience and interpretation	Technical; correctness
Locus of AI	External	External	Platform-internal	Platform-internal
Timing of AI answer	NA	NA	years after posting (treated-group mean pre-policy tenure $\approx$ 49 months)	immediately after posting
AI's relation to the human answerer	None – diverts questions pre-arrival	Collaborator	Supplementary contributor	Direct competitor
AI identifiable	No	No	Yes	Yes
Separates AI <i>label</i> from AI <i>content</i>	No	No	Not the focus	Yes
Research design	Natural experiment	Natural experiment	Natural experiment and lab experiment	Randomized field experiment

Appendix B: Background on the Randomized Experiment and Randomization Check

The platform's randomized experiment appears to be directly linked to the release of ChatGPT. Nine days after ChatGPT's public release on December 9, 2022, SegmentFault implemented a policy prohibiting the posting of AI-generated answers, that is, answers that a user copies from an external tool and submits as the user's own work.¹ It does not, and in practice cannot, prevent users from consulting ChatGPT or learning from it before they post. This decision likely responded to community concerns about reliability, accuracy, and the absence of human verification in AI-generated content. The platform enforced this policy through user reporting and automated detection systems while permitting answers that incorporated human verification and proper citation of AI sources.

Nine months after implementing this policy, the randomized experiment was initiated, likely driven by the intent to explore potential avenues for leveraging the increasingly prevalent and sophisticated GenAI tools. It is plausible that the platform recognized the need to strike a balance between harnessing the capabilities of these technologies and addressing the concerns raised by its community regarding the reliability and credibility of AI-generated content. Through this experimental approach, SegmentFault aims to gather empirical data and user feedback, which will inform its decision-making process and guide the development of strategies for integrating generative AI in a manner that enhances the platform's value proposition while maintaining the integrity and trustworthiness of the community-generated content.

Table B1 presents the balance test results for all pre-determined covariates that could potentially influence treatment assignment. We considered a range of pre-assignment variables, including questioner characteristics and question content features. Questioner characteristics

¹ Although ChatGPT is officially unavailable in China, people in China may access ChatGPT via VPNs. The link for the announcement about the restricted use of external GenAI tools is <https://segmentfault.com/a/1190000043031126>.

include the total number of questions asked, the number of answers provided, account age, overall platform activity near the experiment period, reputation scores, and medal counts (gold, silver, and bronze), which are components of SegmentFault’s user reputation system. Question content features include length (measured by Chinese character count), the presence of multimedia elements (e.g., photos, hyperlinks, tables, and block quotes), LLM-based measures of question characteristics (e.g., linguistic complexity, use of technical jargon, and difficulty scores), and question categories.

To assess the quality of randomization, we both conduct conventional t-tests and calculate standardized differences following Austin’s study (2009). The results in Table B1 demonstrate excellent balance across treatment and control groups. All t-tests yield statistically insignificant results (p-values > 0.05), indicating no significant differences in means between the two groups. Moreover, all standardized differences are well below the commonly accepted threshold of 0.10 (10%), with the absolute values ranging from 0.003 to 0.061. These findings provide strong evidence that our randomization procedure successfully created comparable treatment and control groups, ensuring that any observed treatment effects are not driven by systematic differences in pre-treatment characteristics.

Table B1. Balance Check on Pre-Determined Covariates

Variable	Treatment Group		Control Group		t-test		Austin (2009)’s test	
	Mean	SD	Mean	SD	t-stat	p-value	Standardized Difference	Absolute Std. Diff.
Questioner log reputation score	4.262	1.906	4.351	1.928	-1.405	0.160	-0.047	0.047
number of gold medals earned	3.083	7.412	3.409	8.440	-1.233	0.218	-0.041	0.041
number of silver medals earned	35.365	69.005	38.597	79.258	-1.309	0.191	-0.043	0.043
number of bronze medals earned	94.044	160.497	98.254	170.514	-0.765	0.445	-0.025	0.025
log questioner age (days since registration in the platform)	5.597	2.704	5.583	2.707	0.155	0.877	0.005	0.005
log number of past questions	3.025	2.036	3.082	2.029	-0.842	0.400	-0.028	0.028
log number of past answers	1.746	1.692	1.808	1.756	-1.079	0.281	-0.036	0.036
number of recent 1-month accepted answers	0.287	1.020	0.342	1.214	-1.480	0.139	-0.049	0.049

	number of recent 1-month questions	2.994	4.900	3.186	5.346	-1.126	0.260	-0.037	0.037
	number of recent 1-month answers	0.905	3.131	1.067	4.016	-1.348	0.178	-0.045	0.045
	number of recent 1-month comments	3.055	6.524	3.304	6.532	-1.145	0.252	-0.038	0.038
	number of recent 2-month accepted answers	0.545	1.882	0.673	2.369	-1.798	0.072	-0.060	0.060
	number of recent 2-month questions	5.851	10.229	6.217	11.095	-1.032	0.302	-0.034	0.034
	number of recent 2-month answers	1.636	5.523	1.987	7.503	-1.607	0.108	-0.053	0.053
	number of recent 2-month comments	5.917	11.922	6.563	12.567	-1.584	0.113	-0.053	0.053
	number of recent 3-month accepted answers	0.824	2.849	0.991	3.541	-1.569	0.117	-0.052	0.052
	number of recent 3-month questions	8.871	15.936	9.156	16.819	-0.523	0.601	-0.017	0.017
	number of recent 3-month answers	2.455	7.944	2.855	10.167	-1.321	0.186	-0.044	0.044
	number of recent 3-month comments	9.122	18.062	9.786	19.165	-1.072	0.284	-0.036	0.036
Question	log number of Chinese characters	4.026	1.013	3.959	1.184	1.848	0.065	0.061	0.061
	presence of photos	0.409	0.492	0.415	0.493	-0.358	0.720	-0.012	0.012
	presence of hyperlinks	0.095	0.293	0.087	0.281	0.888	0.374	0.030	0.030
	presence of tables	0.019	0.135	0.017	0.128	0.433	0.665	0.014	0.014
	presence of block quotes	0.004	0.063	0.003	0.052	0.653	0.514	0.022	0.022
	question linguistic complexity (LLM-based)	5.043	1.331	5.047	1.336	-0.092	0.927	-0.003	0.003
	question use of technical jargon (LLM-based)	5.954	1.549	5.901	1.602	1.004	0.315	0.033	0.033
	question difficulty scores (LLM-based)	4.926	1.183	4.931	1.221	-0.108	0.914	-0.004	0.004
	web and mobile development question type	0.576	0.494	0.573	0.495	0.187	0.851	0.006	0.006
	software design and architecture question type	0.010	0.098	0.011	0.106	-0.514	0.607	-0.017	0.017
	cloud computing question type	0.044	0.206	0.053	0.223	-1.171	0.242	-0.039	0.039
	databases question type	0.026	0.159	0.025	0.158	0.114	0.909	0.004	0.004
	computer networks and systems question type	0.012	0.111	0.008	0.090	1.288	0.198	0.043	0.043
	AI and machine learning question type	0.012	0.111	0.010	0.098	0.777	0.438	0.026	0.026
	development tools and environment question type	0.034	0.181	0.029	0.169	0.813	0.416	0.027	0.027
	programming languages question type	0.146	0.353	0.156	0.363	-0.874	0.382	-0.029	0.029
	career and professional development question type	0.005	0.067	0.005	0.073	-0.378	0.705	-0.013	0.013

Note: The raw reputation score in our sample is highly right-skewed, which severely violates the normality assumption underlying the t-test. The log transformation brings the distribution much closer to normality and better satisfies the statistical requirements. Following the suggestion of the reviewer, we conducted a Mann-Whitney-Wilcoxon test to assess balance in reputation scores, which does not assume normality. The result ($W = 1,667,892$, $p = 0.238$) confirms that there is no statistically significant difference in questioner reputation between the treatment

and control groups. For other log-transformed variables, we conducted the same Mann-Whitney-Wilcoxon test, and none of the results was statistically significant (all p-values > 0.40).

Testing balance with individual t-tests across multiple covariates creates a severe multiple comparisons issue. For example, testing 20 covariates at 5% significance would yield one spurious rejection by chance alone even under perfect randomization. While Bonferroni corrections address this, they dramatically reduce statistical power. The regression approach tests joint orthogonality in a single framework, providing a well-calibrated F-test of whether treatment assignment is systematically related to the full set of covariates. We follow Fagereng et al. (2021), who use regression-based balance checks in their randomized lottery study published in the *Journal of Political Economy*, noting that the approach provides both an omnibus test and coefficient-level diagnostics.

Table B2 in the appendix presents the results of our random assignment test. Columns 1 and 2 progressively incorporate questioner characteristics and question content features into multivariate regressions, all including question timing fixed effects. Specifically, to address the high correlation between linguistic complexity and question length, we residualize the complexity measure by regressing it on log question length and extracting the residuals. This approach allows us to isolate the component of linguistic complexity that is orthogonal to question length, thereby avoiding multicollinearity while still testing whether complexity predicts treatment assignment conditional on length. We then include both log question length and the residualized complexity measure in the balance test regressions. We find that none of the pre-assignment characteristics are statistically significant predictors of treatment status. These results provide evidence for the validity of random assignment.

Table B2. Regression-Based Random Assignment Test Results

		(1)	(2)
		DV: treatment status (AI_i)	
<u>Questioner</u>	log reputation score	-0.011	-0.012

		(0.011)	(0.011)
	number of gold medals earned	0.001	0.001
		(0.003)	(0.003)
	number of silver medals earned	-0.001	-0.001
		(0.001)	(0.001)
	number of bronze medals earned	0.000	0.000
		(0.000)	(0.000)
	log questioner age (days since registration in the platform)	0.009	0.008
		(0.006)	(0.006)
	log number of past questions	-0.009	-0.007
		(0.012)	(0.012)
	log number of past answers	0.003	0.001
		(0.010)	(0.010)
	number of recent 1-month accepted answers	0.004	0.003
		(0.029)	(0.029)
	number of recent 1-month questions	0.002	0.002
		(0.006)	(0.006)
	number of recent 1-month answers	-0.003	-0.003
		(0.008)	(0.008)
	number of recent 1-month comments	0.003	0.003
		(0.004)	(0.004)
	number of recent 2-month accepted answers	0.017	0.017
		(0.031)	(0.032)
	number of recent 2-month questions	-0.008	-0.008
		(0.006)	(0.006)
	number of recent 2-month answers	-0.009	-0.009
		(0.010)	(0.010)
	number of recent 2-month comments	-0.003	-0.003
		(0.004)	(0.004)
	number of recent 3-month accepted answers	-0.018	-0.018
		(0.017)	(0.017)
	number of recent 3-month questions	0.005	0.005
		(0.003)	(0.003)
	number of recent 3-month answers	0.008	0.008
		(0.006)	(0.006)
	number of recent 3-month comments	0.001	0.001
		(0.002)	(0.002)
Question	log number of Chinese characters		0.011
			(0.010)
	presence of photos		-0.007
			(0.018)
	presence of hyperlinks		0.025
			(0.030)
	presence of tables		0.014
			(0.070)
	presence of block quotes		0.129
			(0.144)
	question linguistic complexity (LLM-based, residualized)		-0.012
			(0.013)
	question use of technical jargon (LLM-based)		0.011
			(0.008)
	question difficulty scores (LLM-based)		-0.007
			(0.012)
	web and mobile development question type		-0.010

		(0.026)
software design and architecture question type		-0.063
		(0.084)
cloud computing question type		-0.062
		(0.046)
databases question type		0.002
		(0.058)
computer networks and systems question type		0.061
		(0.091)
AI and machine learning question type		0.075
		(0.087)
development tools and environment question type		0.031
		(0.055)
programming languages question type		-0.029
		(0.033)
career and professional development question type		-0.015
		(0.122)
question timing FE	√	√
Num. obs.	3,613	3,613

Note. Standard errors in parentheses. . p<0.1, * p<0.05, ** p<0.01, *** p<0.001.

The randomization was conducted without blocking or stratification according to SegmentFault. Also, through analysis, we verified that SegmentFault has not employed block randomization. Our investigation verified three potential blocking variables:

Questioner-level blocking / topic-based blocking. We tested whether questions from the same user were systematically assigned to the same condition. Our analysis of unique questioners who posted at least two questions during the experimental period shows that 78% had questions in both treatment and control groups, consistent with independent assignment rather than user-level blocking. Similar patterns appear when we tested potential topic-based blocking. Across question tags with at least two questions, 74% had questions in both treatment and control groups.

Temporal blocking. We examined whether treatment assignment varied systematically over time (e.g., alternating hours, days, or weeks). Figure B shows that the daily number of treated questions fluctuates randomly around 50% throughout the experimental period, with a standard deviation of 0.131, consistent with daily independent randomization rather than

temporal blocks. This figure demonstrates the consistent implementation of the AI throughout the full observation window.

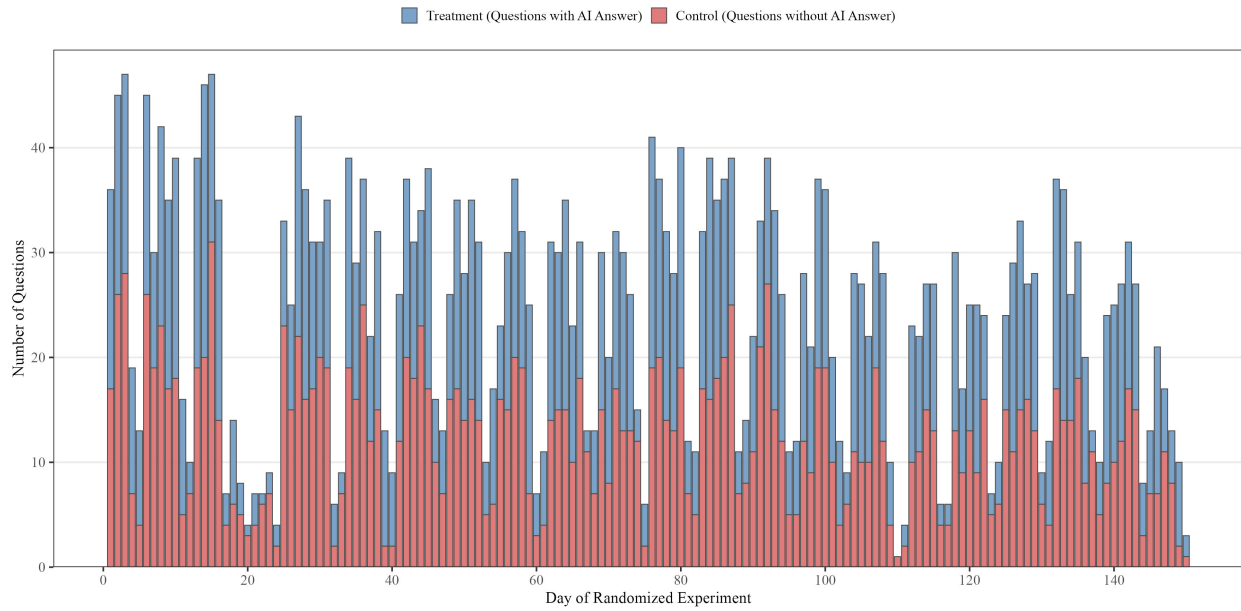


Figure B. The Daily Number of Questions

Appendix C: LLM-Enabled Measures

In designing prompts to assess the content quality of answers and question characteristics (i.e., linguistic complexity, use of technical jargon, and difficulty scores) in an online Q&A community, we employed a comprehensive approach informed by best practices in prompt engineering, particularly drawing from the OpenAI handbook.² Our methodology incorporated several key strategies to enhance the reliability and effectiveness of the assessment process.

Central to our approach was an emphasis on clear instructions. We provided detailed queries to elicit relevant and focused responses from the language model. Specifically, we used Kimi, a prominent LLM developed by Moonshot, a leading vendor in China, to evaluate the overall content quality. To mitigate potential misinterpretations, we conducted thorough validations of the prompts. We implemented a well-crafted system message to establish a specific persona for the model, ensuring consistency in its role throughout the assessment. To further structure the input and output, we employed clear delimiters and specified the desired output format. Additionally, we incorporated a chain-of-thought approach to delineate the steps required to complete the assessment task. This not only made the evaluation process more transparent but also allowed for a more nuanced analysis of the model’s reasoning. To maintain concision and focus, we requested that refined reasons for outputs be provided in single sentences.

A crucial aspect of our prompt design was giving the model sufficient time to “think.” We instructed the language model to generate its own solutions before evaluating the provided answers. Furthermore, we employed techniques such as inner monologue or sequential queries to ensure that the model’s reasoning process remained hidden, preventing the output of unnecessary

² <https://platform.openai.com/docs/guides/prompt-engineering/prompt-engineering>.

content. Our assessment framework comprised two main components: the system prompt and the user prompt. The system prompt defined the role and behavior of the language model. The user prompt contained specific task instructions and the content for evaluation, directing the model's engagement with the material at hand. The prompts used for the assessment task are detailed below:

System:

Use the following step-by-step instructions to evaluate the technical question, measuring how well it solves a problem objectively without any emotion.

****Step 1****: The user will ask for evaluating the objective answer quality. First work out your own answer to the problem. Don't rely on the provided answer as it may not be correct. ****Do not output all your work for this step****.

****Step 2****: Compare your answer to the provided answer. Objectively evaluate the quality of the provided answer by considering the following dimensions: clarity, readability, accuracy, relevance, level of detail, personal experiences, personal insights, innovative approaches, alternative solutions, and engaging storytelling. ****Do not output all your work for this step****.

****Dimensions Definitions****:

1. ****Clarity****: The ability to convey information in a way that is easily understandable, avoiding ambiguity and confusion.
2. ****Readability****: The quality of being easy and pleasant to read, based on factors such as sentence structure, word choice, and answer formatting.
3. ****Accuracy****: The state of being precise, correct, and in conformity with fact or truth, without errors or inaccuracies.
4. ****Relevance****: The degree to which the provided information is applicable, connected, and pertinent to the specific topic or question at hand.
5. ****Level of detail****: The amount and specificity of information provided, ranging from a high-level overview to an in-depth explanation with supporting facts and examples.
6. ****Personal experiences****: The incorporation of personal anecdotes or real-world examples that are specific to the individual answerer and not commonly known or shared.
7. ****Personal insights****: The ability to provide personal opinions, preferences, and recommendations.
8. ****Innovative approaches****: The suggestion of creative, original, or unconventional ideas, solutions, or methodologies that deviate from traditional or widely accepted norms in problem-solving.
9. ****Alternative solutions****: The ability to suggest alternative approaches or solutions that may not be directly related to the original question but can still help solve the underlying problem.
10. ****Engaging storytelling****: The ability to present information or ideas in a compelling, narrative form that captures the reader's attention, makes the content more memorable, and enhances understanding or relatability.

****Step 3****: Reply with an overall quality rating and quality rating for each dimension ****in required format shown below****. Each score should be an integer from 1 to 10, with 1 denoting extremely low quality and 10 denoting extremely high quality. ****After finishing all scorings****, explain the basis for scores of the 10 dimensions within ****one sentence****.

****Required Output Format**:**

Overall quality: 1 to 10

Clarity: 1 to 10

Readability: 1 to 10

Accuracy: 1 to 10

Relevance: 1 to 10

Level of detail: 1 to 10

Personal experiences: 1 to 10

Personal insights: 1 to 10

Innovative approaches: 1 to 10

Alternative solutions: 1 to 10

Engaging storytelling: 1 to 10

Basis for scores: briefly explain the basis for the overall quality score and 10 scores of dimensions here in one sentence

User:

You need to objectively evaluate the quality of the given answer based on the following dimensions: clarity, readability, accuracy, relevance, level of detail, personal experiences, personal insights, innovative approaches, alternative solutions, and engaging storytelling. Provide a quality score from 1-10 for overall quality and then each dimension**in required format shown below**, with 1 being extremely low quality and 10 being extremely high quality.**After finishing all scorings**, explain the basis for scores of the 10 dimensions within**one sentence**.

Here is the**required Output Format**:

Overall quality: 1 to 10

Clarity: 1 to 10

Readability: 1 to 10

Accuracy: 1 to 10

Relevance: 1 to 10

Level of detail: 1 to 10

Personal experiences: 1 to 10

Personal insights: 1 to 10

Innovative approaches: 1 to 10

Alternative solutions: 1 to 10

Engaging storytelling: 1 to 10

Basis for scores: briefly explain the basis for scores of the 10 dimensions within one sentence

We conducted several tests to validate the LLM-enabled content quality ratings. First, we examined cases where the internal GenAI was unable to provide a substantive answer, typically responding with a statement like “I’m sorry, but I can’t give a detailed answer because of [reason].” If the LLM-enabled content quality ratings were valid, these failed answers should receive lower ratings compared to other answers. A t-test confirmed that failed AI answers indeed received significantly lower ratings than other answers ($t = -3.889$, $p = 0.001$).

Second, we compared the LLM-enabled content quality ratings between human answers accepted by questioners and those not accepted. We expected accepted human answers to receive higher ratings. This expectation was supported by a t-test, which showed that LLM-enabled content quality ratings for accepted human answers were significantly higher than those for non-accepted answers ($t = 2.006$, $p = 0.045$).

Third, we investigated the relationship between LLM-enabled content quality ratings and recognition by viewers. Specifically, we used net likes to quantify recognition by viewers and divided answers into two groups based on the median net likes. Comparing these two groups, we found that answers with above-median net likes consistently received higher LLM-enabled content quality ratings. This difference was statistically significant ($t = 3.235$, $p = 0.001$). This result further supports the validity of the LLM-enabled content quality ratings. Taken together, these tests provide strong evidence for the validity and reliability of the LLM-enabled content quality ratings.

Similarly, we measure the linguistic complexity, use of technical jargon, and question difficulty scores. The prompt for each metric is as follows:

System:

Use the following step-by-step instructions to evaluate the technical question, measuring the target metric objectively without any emotion.

Target metric: {metric}

****Step 1**:** Read the question carefully. Identify its topic, length, phrasing, and any domain-specific markers.

****Step 2**:** Independently assess the target metric for this question only. Do not answer the question. Do not evaluate any answer. Focus strictly on measuring the question's {metric}.

****Step 3**:** Provide a single overall rating using a scale ranging from 1 to 10 (integer), where 1 denotes extremely low and 10 denotes extremely high levels of the target metric. Then briefly explain the key factors considered.

Your reply must include:

- A single overall rating (an integer 1-10).
- A short explanation (a single-sentence).

User:

You need to evaluate the objective level of this question on a single target metric, without any emotion or subjective preference. Do NOT answer the question. Do NOT evaluate any provided answers. Reply only with a JSON object that contains an ****integer score from 1-10 for the target metric, and a single-sentence scoring basis.****

Output format (strict JSON, no extra text):\n'

{ "\n"

"metric": "<insert the target metric name exactly>",\n'

"score": <insert integer 1-10>,\n'

"basis": "<insert a single concise sentence explaining the rating>"\n'

"} "

Appendix D: Partial Identification of the AI Quality Effect Using Lee Bounds

Bounding the Effect of AI Quality Under Sample Selection. We are interested in the slope coefficient α_2 from the model

$$Y = \alpha_0 + \alpha_1 D + \alpha_2 (D \times Q) + \varepsilon,$$

where D is a binary indicator for treatment (the presence of an AI answer), Q is a continuous measure of AI quality (defined only when $D = 1$), and Y is the outcome of interest (the quality of the first human answer). The outcome Y is observed only for questions that receive at least one human answer. We denote this by $S = 1$ when a question receives at least one human answer and $S = 0$ when it receives none, so that Y is observed if and only if $S = 1$.

If the questions that attract a human answer differ systematically from those that do not, the estimate of α_2 can be biased. We therefore adopt the partial identification approach of Lee (2009), which provides sharp bounds on the average treatment effect (ATE) for a binary treatment under a monotonicity assumption on selection, and we adapt it to bound the continuous-treatment slope α_2 . Because our adaptation divides binary bounds by a difference in average quality and combines bounds across groups, the resulting interval for α_2 is valid and conservative rather than sharp.

Lee’s method is designed for a binary treatment, whereas AI quality is continuous. Let T denote a generic binary treatment used only within the two strategies below. In each strategy, T is a binarized transformation of AI quality Q , and it is therefore distinct from the AI answer-presence indicator D . Lee’s method assumes that the potential selection indicator satisfies $S(1) \geq S(0)$ almost surely, that is, receiving T does not reduce the probability of being observed. Under this assumption, the observed outcome distribution for the $T = 1$ group is a mixture of individuals who would be observed regardless of T and individuals who are observed

only when $T = 1$. By trimming the $T = 1$ group's outcome distribution by the fraction $1 - p_0/p_1$, where $p_1 = P(S = 1 | T = 1)$ and $p_0 = P(S = 1 | T = 0)$, one obtains sharp bounds on the ATE for the always-observed subpopulation. In our setting, we treat all missingness arising from the focus on the first human answer as a form of selection, and we apply the bounds to a complete-observation indicator that equals one when a human answer is present and its quality is recorded. This is a conservative choice that widens the bounds but requires no additional assumptions about the missingness mechanism (Manski 2003). If monotonicity is violated in the direction $p_1 < p_0$, we follow common practice and swap the treatment and control labels, then reverse the sign of the resulting bounds (Lee 2009).

We implement two complementary strategies to translate this binary framework into bounds on the slope α_2 .

Approach A: Median split within the treated sample. We restrict attention to questions with an AI answer ($D = 1$) and split them at the median of Q into a low-quality group ($T = 0$) and a high-quality group ($T = 1$). Applying the Lee bounds to this binary treatment yields an interval $[L_A, U_A]$ for the average difference in Y between the high- and low-quality groups. Under the linear model, this difference equals $\alpha_2 \times (\bar{Q}_{\text{high}} - \bar{Q}_{\text{low}})$. Scaling the bounds by the difference in average quality therefore gives a valid interval for the effect of AI answer quality:

$$\alpha_2 \in \left[\frac{L_A}{\bar{Q}_{\text{high}} - \bar{Q}_{\text{low}}}, \frac{U_A}{\bar{Q}_{\text{high}} - \bar{Q}_{\text{low}}} \right].$$

This approach is simple and directly applicable, but it uses only a coarse dichotomization of Q and therefore may be less efficient than using the full continuous variation.

Approach B: Tercile Bins Compared with the Control Group. To exploit more of the variation in Q , we divide the treated questions into terciles of Q . For each tercile, we define $T = 1$ for questions in that tercile (with $D = 1$) and $T = 0$ for all control questions ($D = 0$), and we

compute Lee bounds on the ATE of belonging to that tercile relative to the control group, using the ratio of observation probabilities to set the trimming fraction. If a tercile has a higher observation rate than the control group, we swap the groups and reverse the sign, as described above. This yields, for each tercile, an interval for the ATE evaluated at that tercile’s average quality level.

Because α_2 is the marginal effect of Q , we can bound it using the **extreme bins**: those with the lowest and highest average Q . Let Q_{low} and Q_{high} be the mean Q in the lowest and highest terciles, and let $[L_{\text{low}}, U_{\text{low}}]$ and $[L_{\text{high}}, U_{\text{high}}]$ be their respective ATE bounds. Any true ATE values lying within these intervals imply a slope of $(ATE_{\text{high}} - ATE_{\text{low}})/(Q_{\text{high}} - Q_{\text{low}})$. Therefore, the most conservative interval for α_2 consistent with all possible realizations is

$$\alpha_2^{\text{lower}} = \frac{L_{\text{high}} - U_{\text{low}}}{Q_{\text{high}} - Q_{\text{low}}}, \alpha_2^{\text{upper}} = \frac{U_{\text{high}} - L_{\text{low}}}{Q_{\text{high}} - Q_{\text{low}}}.$$

This interval is valid without imposing any functional form on the relation between Q and the ATE beyond that implied by the linear model, and it follows the logic of interval arithmetic in partial identification (Imbens & Manski 2004; Manski 2003). If some terciles violate monotonicity even after swapping, they are excluded, potentially restricting the range of Q used for identification.

Results. Applying Approach A, we obtain bounds on the effect of AI answer quality [0.047, 0.402]. Approach B yields a slightly wider but largely overlapping interval of [0.042, 0.337]. Both intervals are strictly positive and exclude zero, providing robust evidence that higher AI quality is associated with a higher outcome among treated questions, even after accounting for non-random sample selection. The intervals are relatively wide, reflecting the conservative nature of the Lee bounds, but they are mutually consistent in sign and magnitude.

The minor differences between the two approaches stem from the different ways in which the continuous quality measure is discretized and the resulting leverage of extreme quality values.

Conclusion. The Lee bounds indicate that a one-unit increase in AI quality is associated with an approximately 0.04-0.40 unit increase in the first human answer quality, with both approaches producing overlapping intervals that exclude zero. This suggests a statistically meaningful and economically significant positive effect of AI quality among treated questions, robust to non-ignorable selection into observed responses. The consistency of results across two distinct bounding strategies strengthens confidence in the qualitative conclusion.

Appendix E: Overview of Main Analyses: Sample Sizes and Selection Criteria

In our paper, different analyses correspond to different samples used. To eliminate any ambiguity, we include a table below that clearly delineates the major analyses conducted, their corresponding sample sizes, and the specific sample selection criteria.

Table E. Overview of Main Analyses: Sample Sizes and Selection Criteria

Table or analysis	Purpose	Sample size	Selection criteria and reason for differences
Table 3, Column 1	Average effect on human answer supply	3,613	Full question-level experimental sample.
Table 3, Column 2	AI label, length, and quality effects on human answer supply	3,612	Excludes one treated question because the LLM declined to provide the assessment for a specific AI answer.
Table 3, Column 3	Average effect on first human answer quality	2,727	Questions with at least one human answer and a usable first human answer quality outcome.
Table 3, Column 4	Moderated effects on first human answer quality	2,726	Uses the Column 3 sample and additionally requires usable AI answer quality score; one observation is excluded for the LLM assessment issue above.
Table 5, Columns 1-4	Alternative models for human answer supply	3,612 each	Same complete AI-profile sample as Table 3, Column 2.
Table 5, Column 5	Questioner-random-effect model for first human answer quality	2,726	Same sample as Table 3, Column 4.
Table 6, Columns 1-3	Alternative answer-count windows	3,612 each	Same complete AI-profile sample; the outcome window changes to 3 days, 14 days, or the full observation period.
Table 6, Columns 4-6	Early experiment-period cohorts for human answer supply	837 / 1,606 / 2,412	Questions posted during the first 30, 60, or 90 days after launch.
Table 7, Columns 1-3	Early experiment-period cohorts for first human answer quality	592 / 1,214 / 1,833	Corresponding 30-, 60-, and 90-day posting cohorts, subject to the model's first-human-answer and measurement requirements.
Table 8	Historical spillover comparison	5,984	1,847 experimental-period control questions plus 4,137 historical questions from the pre-AI period.
Table 9	Oster sensitivity analysis for human answer supply	3,613	Full sample used for the average answer-supply estimate in Table 3, Column 1.
Table 10	Question-difficulty heterogeneity	2,377 difficult / 1,235 easy	Mean split of the 3,612-observation complete AI-profile sample. Splitting at the median produces similar results.
Table 11, Columns 1-3	Coverage mechanism for human answer supply	3,613 / 3,612 / 3,613	Column 2 requires AI answer quality and excludes the one unassessable observation; Columns 1 and 3 do not require that score.
Table 12, Column 1	First human answer accuracy	2,728	Analysis-specific full sample for the accuracy specification.
Table 12, Column 2	First human contributor expertise	2,747	Largest usable sample for the first-human-contributor expertise outcome.
Table 13	AI-to-first-human-answer similarity	1,277	Questions with at least one human answer in treatment group.

REFERENCES

- Austin, P. C. (2009). Balance Diagnostics for Comparing the Distribution of Baseline Covariates between Treatment Groups in Propensity-score Matched Samples. *Statistics in Medicine*, 28(25), 3083–3107.
- Fagereng, A., Mogstad, M., & Rønning, M. (2021). Why Do Wealthy Parents Have Wealthy Children? *Journal of Political Economy*, 129(3), 703–756.
- Imbens, G. W., & Manski, C. F. (2004). Confidence Intervals for Partially Identified Parameters. *Econometrica*, 72(6), 1845–1857.
- Lee, D. S. (2009). Training, Wages, and Sample Selection: Estimating Sharp Bounds on Treatment Effects. *The Review of Economic Studies*, 76(3), 1071–1102.
- Manski, C. F. (2003). *Partial Identification of Probability Distributions*.
- Shan, G., & Qiu, L. (2026). Examining the Impact of Generative AI on Users' Voluntary Knowledge Contribution: Evidence from a Natural Experiment on Stack Overflow. *Information Systems Research*, 37(2), 1021–1041.
- Su, Y., Zhang, K., Wang, Q., & Qiu, L. (2025). *Generative AI and Human Knowledge Sharing: Evidence from A Natural Experiment*. SSRN Scholarly Paper.
- Xue, J., Wang, L., Zheng, J., Li, Y., & Tan, Y. (2026). Can ChatGPT Kill User-Generated Q&A Platforms? *Information Systems Research*.